

# Hash Tables: Hash Functions, Sets & Maps

Hash Table →

|   |      |
|---|------|
| 0 | gre  |
| 1 |      |
| 2 | greg |
| 3 |      |
| 4 |      |

linked list [separate chaining]

'greg'  
we need to hash it. to put it in the hash table. → we need to apply a hash function.

greg

String (Input) → Index (Output)

g: 7

r: 5

e: 18

"greg"

Hash function (eg) → add then % by 5  
 $75187 = 37 \% 5 = 2$

gre

 $7518 = 30 \% 5$  $= 0$ 

grg

 $7187 = 32 \% 2$  $= 2$ 

collision problem

To solve collision problem

Separate chaining

Linear probing

The main purpose of hash table to exist is to implement hash sets & hash maps

Hash Sets: A collection of unique items

- looking up an element in hash table takes  $O(1)$
- add  $\rightarrow O(1)$ ; delete  $\rightarrow O(1)$ ; lookup  $\rightarrow O(1)$  (amortized)

Hash Maps: All the functionalities of hash set, except it can store data

keys must be hashable

key : value

greg : 7

 $37 \% 5 = 2$ 

gre : 5

 $30 \% 5 = 2$ 

|   |           |
|---|-----------|
| 0 | (gre, 5)  |
| 1 |           |
| 2 | (greg, 7) |
| 3 |           |
| 4 |           |

Lookup  $* O(1)$ 

return if key exist or not (True / False) OR return the value associated with the key

Add  $O(1)$ Remove  $* O(1)$

Linear Probing

g: 7

e: 5

r: 18

greg  $\Rightarrow 32\% \cdot 5 = 2$ 

|   |      |
|---|------|
| 0 | greg |
| 1 | gr   |
| 2 | greg |
| 3 | grg  |
| 4 |      |

full  
check next oneempty greg  $\Rightarrow 37\% \cdot 5 = 2$ full gte  $\Rightarrow 30\% \cdot 5 = 0$ empty gr  $\Rightarrow 25\% \cdot 5 = 0$ 

collision

In this case we don't change,  
instead we check for another spot in the  
hash table

lookup  $*O(1)$ Delete  $O(1)$ 

if we hit a blank, then the element doesn't exist

if we remove something then replace it with something else like -1 so we know we had something there and our search can continue when looking up through linear probing.

Delete  $O(1)$ What is Hashable?(Immutable)  $\rightarrow$  Strings $\rightarrow$  Integers $\rightarrow$  TuplesWhat is Not Hashable? $\rightarrow$  Arrays (Mutable) $\rightarrow$  Dictionaries

Python specific

Python

 $\rightarrow$  set $\rightarrow$  dictionary

$\rightarrow$  defaultdict from  
collections  
import

 $\rightarrow$  Counter